



SAPIENZA
UNIVERSITÀ DI ROMA

DEPARTMENT OF MECHANICAL AND AEROSPACE ENGINEERING

Homework 3 — Attitude Control of a Launch Vehicle in Atmospheric Flight

DYNAMICS AND CONTROL OF LAUNCH VEHICLES

Professor:

Prof. A. Zavoli

Student:

Niccolò D'Ambrosio

Academic Year 2025/2026

Contents

1	Introduction	2
1.1	Plant model	2
1.2	Control architecture	3
1.3	A note on conditional stability	3
2	Task 1 — Rigid Launch Vehicle	4
2.1	Rigid model and loop	4
2.2	Tuning	4
2.3	Wind-gust response	5
3	Task 2 — Full Launch Vehicle Model	7
3.1	Step B — actuator, delay and bending without compensation	7
3.2	Step C — bending notch	7
3.3	Comparison with the rigid model	8
4	Task 3 — Robustness to Parametric Uncertainty	9
4.1	Frequency- and time-domain analysis	9
4.2	Discussion	9
5	Conclusions	11
	References	12

1 Introduction

This report addresses the design of an attitude control system for a launch vehicle (LV) during the atmospheric phase of flight, at the instant of maximum dynamic pressure ($t = 72$ s, $\max\bar{q}$). At this condition the LV is the hardest to control: the aerodynamic moment destabilises the airframe while the wind disturbance is largest. The design is carried out entirely with *classical* frequency-domain tools, using the Nichols chart as the primary stability indicator, following the short-period pitch-plane model of Greensite [1].

1.1 Plant model

The pitch-plane short-period dynamics are described by the linear time-invariant model (assignment Eq. 1), with state vector $\mathbf{x} = [z, \dot{z}, \theta, \dot{\theta}, \eta, \dot{\eta}]^\top$ (lateral drift, pitch attitude and first bending mode), control input δ (TVC deflection) and disturbance α_w (wind angle of attack):

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & a_1 & a_1 V + a_4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & A_6/V & A_6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\omega_1^2 & -2\zeta_1\omega_1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ a_3 \\ 0 \\ K_1 \\ 0 \\ -\phi_{1,TVC}T_c \end{bmatrix} \delta + \begin{bmatrix} 0 \\ -a_1 V \\ 0 \\ -A_6 \\ 0 \\ 0 \end{bmatrix} \alpha_w. \quad (1)$$

The pitch channel $\ddot{\theta} = A_6 \theta + K_1 \delta - A_6 \alpha_w$ is the core of the problem. The coefficient $A_6 = N_\alpha l_\alpha / I_{yy} > 0$ (*aerodynamic moment*, denoted μ_α) places an open-loop pole at $s = +\sqrt{A_6} = +1.84$ rad/s in the right half-plane: the airframe is **statically unstable** and feedback is mandatory. The coefficient $K_1 = T_c l_c / I_{yy}$ (*control effectiveness*, μ_c) sets the authority of the TVC.

The sensed quantities are produced by the inertial unit (assignment Eq. 2), which couples the bending coordinate η into the gyro and accelerometer channels through $\sigma_{1,ins}$ and $\phi_{1,ins}$:

$$\theta_m = \theta + \sigma_{1,ins} \eta, \quad \dot{\theta}_m = \dot{\theta} + \sigma_{1,ins} \dot{\eta}, \quad z_m = z - \phi_{1,ins} \eta, \quad \dot{z}_m = \dot{z} - \phi_{1,ins} \dot{\eta}. \quad (2)$$

This contamination is exactly what makes the lightly damped bending mode ($\omega_{BM} = 18.9$ rad/s, $\zeta_{BM} = 0.005$) visible to the controller and drives the need for a dedicated filter (Task 2).

The actuator is the second-order TVC of Eq. 3,

$$W_{TVC}(s) = \frac{\omega_{TVC}^2}{s^2 + 2\zeta_{TVC}\omega_{TVC}s + \omega_{TVC}^2}, \quad \omega_{TVC} = 70 \text{ rad/s}, \quad \zeta_{TVC} = 0.7, \quad (3)$$

in series with a pure transport delay $\tau = 20$ ms, approximated by a third-order Padé rational function. Parameters at $t = 72$ s are read from the reference data

set `GreensiteLPV_DATA.mat` (cross-checked against Table 1 of the assignment); the salient values are reported in Table 1.

Table 1: Pitch-plane LV parameters at the max- $\bar{q}$ condition ($t = 72$ s).

Quantity	Symbol	Value
Aerodynamic moment	$A_6 = \mu_\alpha$	3.382 s^{-2}
Control effectiveness	$K_1 = \mu_c$	4.565 s^{-2}
Drift coefficients	a_1, a_3, a_4	$-0.0154, 20.61, -27.27 \text{ s}^{-2}$
Relative velocity	V	937.7 m/s
Bending mode	ω_{BM}, ζ_{BM}	$18.9 \text{ rad/s}, 0.005$
INS coupling	$\phi_{1,ins}, \sigma_{1,ins}$	$0.8, 0.178$
TVC actuator	$\omega_{TVC}, \zeta_{TVC}, \tau$	$70 \text{ rad/s}, 0.7, 20 \text{ ms}$

1.2 Control architecture

A proportional–derivative attitude law is adopted, augmented by a weak negative feedback on the lateral-drift channel (the *drift-minimum* principle):

$$\delta_{cmd} = K_{P,\theta} (\theta_{ref} - \theta_m) - K_{D,\theta} \dot{\theta}_m - K_{P,z} z_m - K_{D,z} \dot{z}_m, \quad (4)$$

with $K_{P,z}, K_{D,z}$ small and negative, of order 10^{-3} , as recommended in the assignment guidelines. The actual deflection follows from the actuator and the bending filter $H_x(s)$ (Task 2): $\delta = W_{TVC}(s) e^{-\tau s} H_x(s) \delta_{cmd}$.

1.3 A note on conditional stability

Because the open-loop airframe carries a right-half-plane pole, the loop is *conditionally stable*: closed-loop stability is lost if the loop gain is *reduced* below a threshold, not only if it is increased. Consequently the `margin` routine returns the low-frequency *aerodynamic* gain margin, and the assignment targets $|GM| \approx 6 \text{ dB}$ and $|PM| \approx 30^\circ$ are interpreted in magnitude, as read off the Nichols chart. This is standard practice for the attitude control of aerodynamically unstable launchers [2].

2 Task 1 — Rigid Launch Vehicle

2.1 Rigid model and loop

Neglecting the bending mode, the plant reduces to the four states $[z, \dot{z}, \theta, \dot{\theta}]$ and the actuator is taken as ideal ($W_{TVC} = e^{-\tau s} = 1$). The open-loop transfer broken at the TVC deflection, in the $1 + L$ convention, is obtained by closing the static PD law of Eq. (4) around the plant:

$$L(s) = K_{P,\theta} G_{\delta \rightarrow \theta}(s) + K_{D,\theta} G_{\delta \rightarrow \dot{\theta}}(s) + K_{P,z} G_{\delta \rightarrow z}(s) + K_{D,z} G_{\delta \rightarrow \dot{z}}(s), \quad (5)$$

where $G_{\delta \rightarrow \bullet}$ are the plant channels from δ to each measurement. The dominant term is the pitch loop $L_\theta(s) = (K_{P,\theta} + K_{D,\theta}s) K_1 / (s^2 - A_6)$, whose closed-loop characteristic polynomial $s^2 + K_1 K_{D,\theta} s + (K_1 K_{P,\theta} - A_6)$ is stable only for $K_{P,\theta} > A_6 / K_1 = 0.74$ and $K_{D,\theta} > 0$: the proportional gain must overcome the aerodynamic instability.

2.2 Tuning

The pitch gains were tuned (by a small `fminsearch` wrapper around the margin computation) to meet the assignment targets, while the lateral gains were fixed at $K_{P,z} = K_{D,z} = -10^{-3}$. The resulting design is

$$K_{P,\theta} = 1.98, \quad K_{D,\theta} = 1.40, \quad K_{P,z} = K_{D,z} = -10^{-3}, \quad (6)$$

giving $|GM| = 6.0$ dB at 0.34 rad/s and $|PM| = 30^\circ$ at 0.21 rad/s. The Nichols chart of Figure 1 shows the curve threading between the two critical points $(-180^\circ, 0$ dB) and $(-540^\circ, 0$ dB) — the geometric signature of conditional stability — and clearing the 6 dB closed-loop contour.

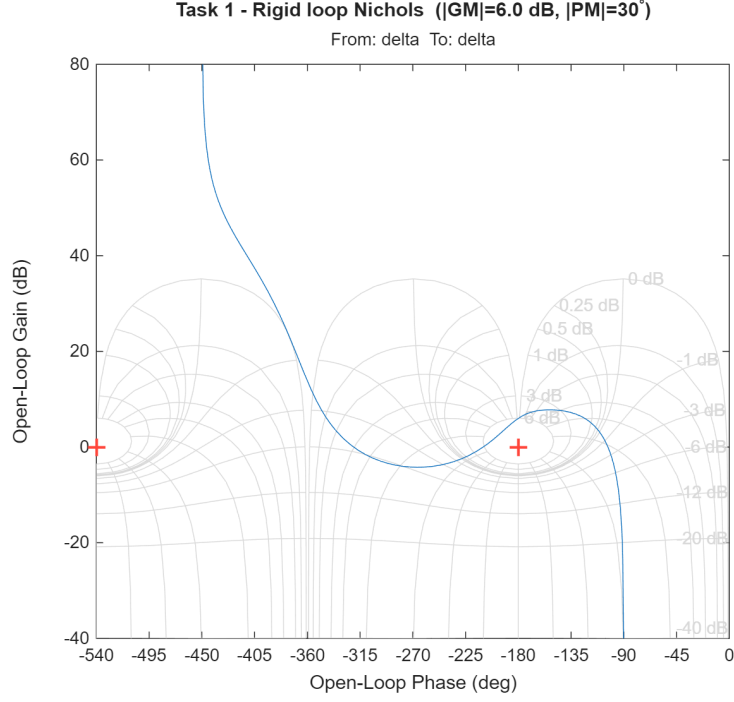


Figure 1: Task 1 — Nichols chart of the rigid open-loop transfer. The loop is conditionally stable; the design meets $|GM| \approx 6$ dB and $|PM| \approx 30^\circ$.

2.3 Wind-gust response

The closed loop is excited by a deterministic “1-cosine” wind gust whose amplitude ($V_g = 6.4$ m/s) is taken from the reference dry-wind dispersion (`drywind.mat`, severe regime) at the current altitude, converted to a wind angle of attack $\alpha_w = v_w/V$. The four key time histories requested by the assignment are shown in Figure 2: the pitch attitude stays below 0.14° , the TVC deflection peaks at 0.38° , and the lateral drift builds up smoothly to a few metres, as expected from the weak drift feedback. The response is well damped and the actuator effort is modest.

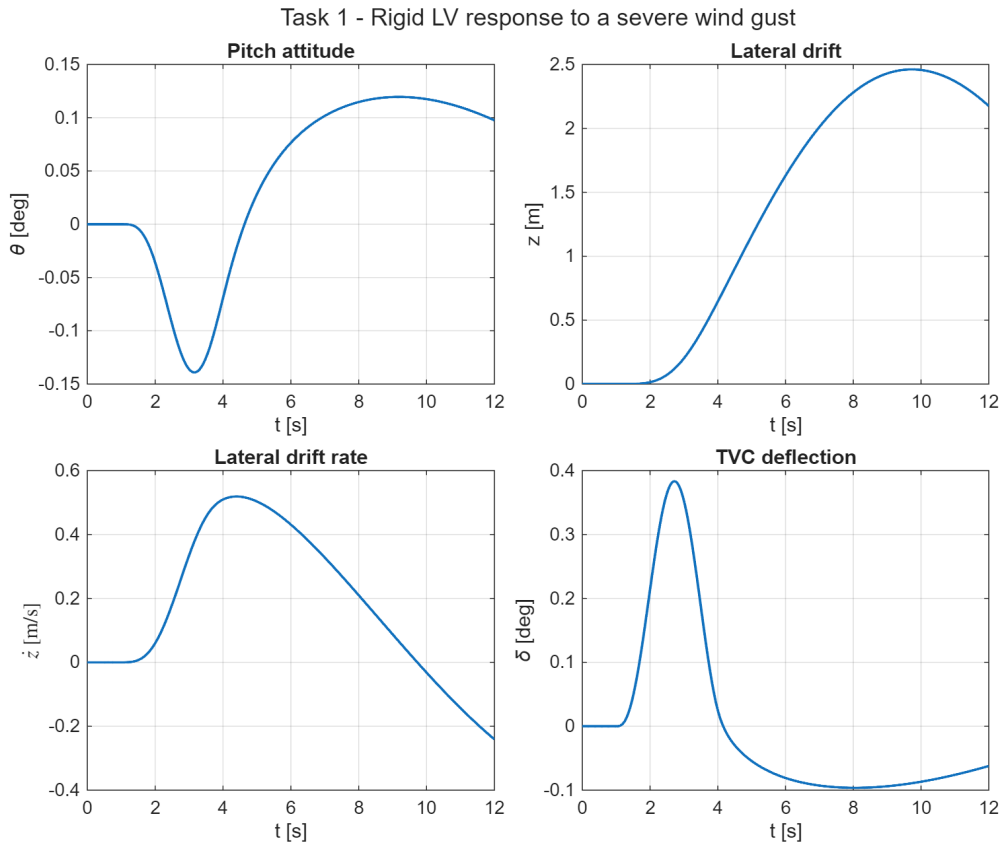


Figure 2: Task 1 — response to a severe wind gust: pitch attitude θ , lateral drift z , drift rate $\dot{z}$ and TVC deflection δ .

3 Task 2 — Full Launch Vehicle Model

Task 2 reintroduces the full dynamics of Eq. (1): the lightly damped first bending mode, the second-order TVC actuator of Eq. (3) with its 20 ms delay, and the INS measurement model of Eq. (2). The design proceeds incrementally, as suggested in the assignment guidelines.

3.1 Step B — actuator, delay and bending without compensation

Adding the TVC, delay and bending mode to the Task 1 controller leaves the low-frequency margins essentially intact, but the bending resonance appears in the loop at a magnitude of $|L(\omega_{BM})| = +39$ dB. With a damping of only $\zeta_{BM} = 0.005$ this peak encircles the critical point and the closed loop becomes **unstable** (a closed-loop pole at $\text{Re}(s) = +5.6$). The red curve in Figure 3 shows the bending loop crossing into the unstable region.

3.2 Step C — bending notch

The guideline filter of Eq. 4 is used. Two strategies are possible: a *phase*-stabilising lead-lag (the non-minimum-phase variant, which shifts the bending excursion left/right on the Nichols chart) and a *gain*-stabilising notch (a minimum-phase band-reject section that drives the resonance below 0 dB). The lead-lag was found to thin the bending phase margin to a fraction of a degree, whereas the gain-stabilising notch

$$H_x(s) = \frac{s^2 + 2\zeta_{x,N}\omega_x s + \omega_x^2}{s^2 + 2\zeta_{x,D}\omega_x s + \omega_x^2}, \quad \omega_x = \omega_{BM}, \quad \zeta_{x,N} = 0.002, \quad \zeta_{x,D} = 0.7, \quad (7)$$

attenuates $|L(\omega_{BM})|$ to -12 dB while remaining flat at the low-frequency rigid crossovers. The bending mode is therefore left at its open-loop (lightly damped but stable) location, the rigid-body gain margin is *preserved* ($|GM| = 6.6$ dB) and the phase margin is $|PM| = 26.5^\circ$ (a few degrees below the 30° of the ideal-actuator rigid loop of Task 1, the difference being the phase lag added by the second-order TVC and the 20 ms transport delay), and the closed loop is stable. No re-tuning of the PD gains was necessary, consistent with the guideline note. The notch is the chosen design; the blue curve in Figure 3 shows the stabilised loop.

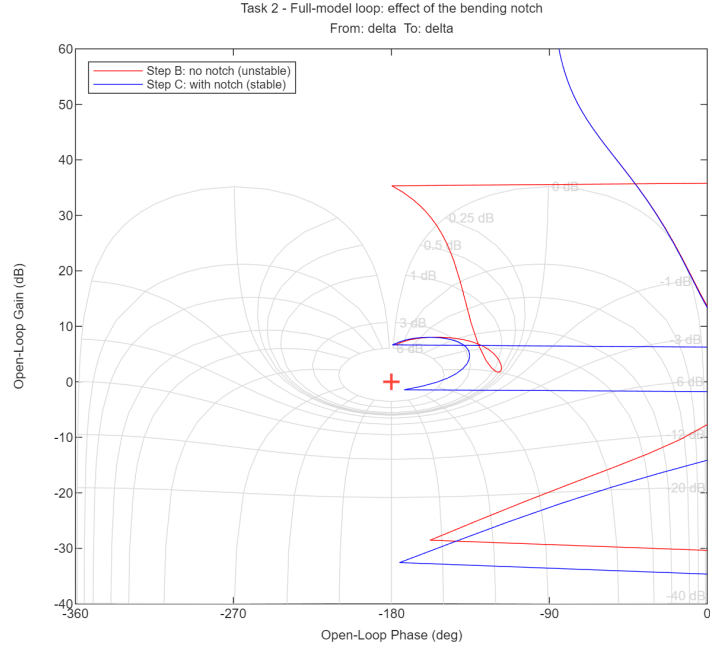


Figure 3: Task 2 — effect of the bending notch on the full-model loop: without the notch (red) the +39 dB bending resonance destabilises the loop; with the notch (blue) the resonance is gain-stabilised and the rigid margins are kept.

3.3 Comparison with the rigid model

Because the notch is localised at the bending frequency and the TVC bandwidth (70 rad/s) is well above the control crossover, the full-model gust response is virtually indistinguishable from the rigid one (Figure 4): the pitch peak changes from 0.139° to 0.132° and the TVC effort is unchanged. The added dynamics affect stability robustness (the bending margin) far more than nominal performance.

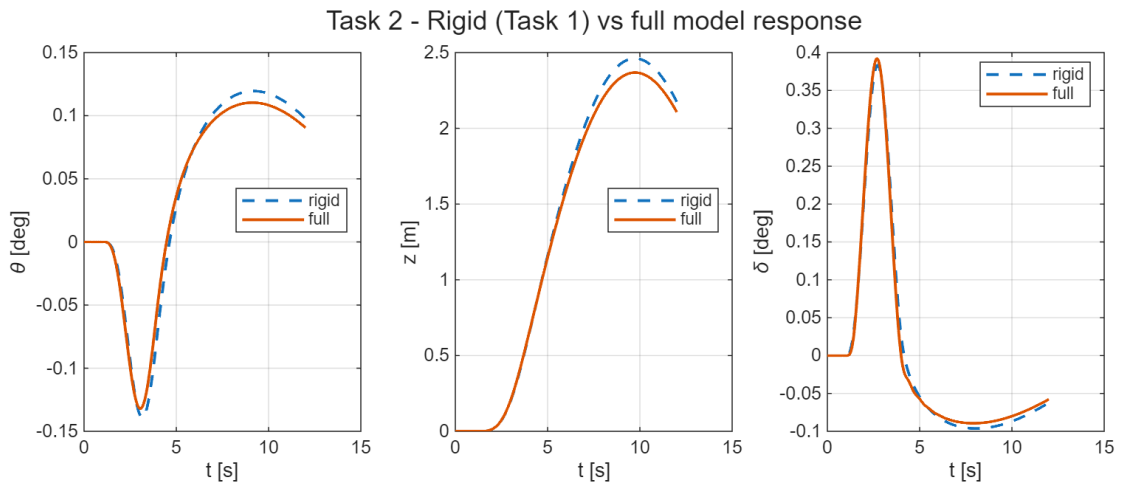


Figure 4: Task 2 — rigid (Task 1) versus full-model gust response: pitch attitude, lateral drift and TVC deflection are nearly coincident.

4 Task 3 — Robustness to Parametric Uncertainty

The aerodynamic moment $\mu_\alpha = A_6$ and the control effectiveness $\mu_c = K_1$ are treated as uncertain. The controller designed in Task 2 (PD gains plus bending notch) is held **fixed** and its robustness is assessed over the four corner cases obtained by varying μ_α and μ_c independently by $\pm 30\%$, together with the nominal case.

4.1 Frequency- and time-domain analysis

Table 2 collects, for each corner, the rigid-body (low-frequency) gain margin, the smallest gain margin over the whole loop (which is governed by the bending shoulder), the phase margin, and the peak pitch and lateral-drift excursions to the same severe gust. *All four corners remain stable.*

Table 2: Task 3 — margins and gust-response peaks over the $\pm 30\%$ corners.

Case	μ_α	μ_c	rigid $ GM $ [dB]	min $ GM $ [dB]	$ PM $ [°]	peak θ [°]	peak z [m]
Nominal	1.00	1.00	6.58	1.81	26.5	0.132	2.37
C1	0.70	1.00	9.44	1.79	25.7	0.085	1.91
C2	1.30	1.00	4.47	1.83	22.1	0.187	2.88
C3	1.00	0.70	3.59	2.82	18.6	0.219	3.16
C4	1.00	1.30	8.79	0.91	11.8	0.093	2.01

The Nichols overlay (Figure 5) shows the corners clustering tightly at low frequency and fanning out near the bending frequency; the time-domain overlay (Figure 6) shows the spread of the gust response.

4.2 Discussion

The two uncertainties stress different parts of the loop:

- A larger aerodynamic moment ($\mu_\alpha \uparrow$, case C2) makes the airframe *more* unstable (the RHP pole moves further right), which erodes the rigid-body gain margin (from 6.6 to 4.5 dB) and produces the largest pitch and drift excursions together with the low- μ_c case.
- A larger control effectiveness ($\mu_c \uparrow$, case C4) raises the overall loop gain by ≈ 2.3 dB; this *improves* the rigid margin but pushes the bending resonance closer to instability, collapsing the minimum gain margin to 0.9 dB. The bending channel, not the rigid one, is the binding constraint in this corner.
- The benign corners are C1 ($\mu_\alpha \downarrow$, a less unstable airframe) and, for the bending margin, C3.

In summary, performance is worst for high aerodynamic instability while stability robustness is tightest for high control authority; nevertheless the fixed Task 2 controller tolerates the entire $\pm 30\%$ uncertainty box without re-design, confirming the adequacy of the classical design at the max- $\bar{q}$ condition.

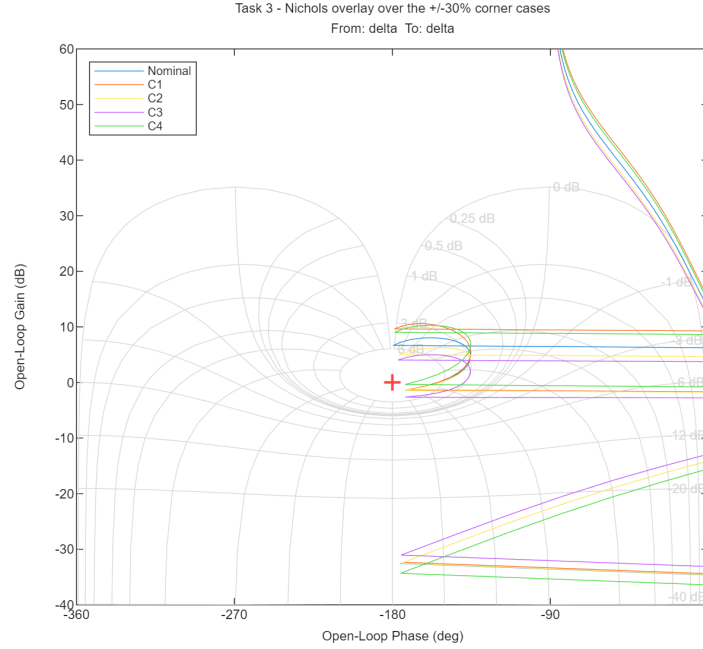


Figure 5: Task 3 — Nichols overlay over the nominal and four corner cases.

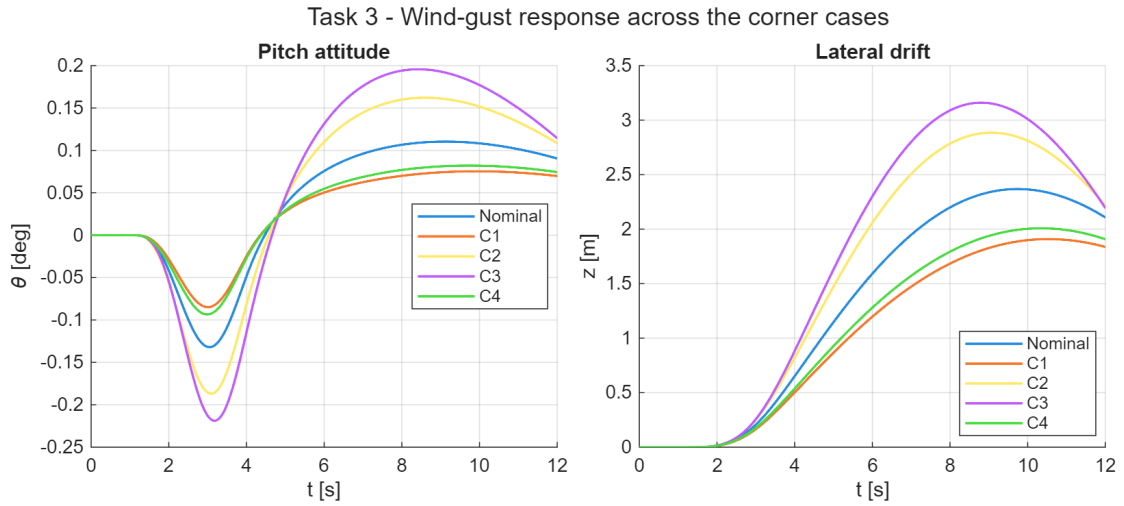


Figure 6: Task 3 — wind-gust response across the corner cases: pitch attitude (left) and lateral drift (right).

5 Conclusions

A classical attitude control system for an aerodynamically unstable launch vehicle at the max- $\bar{q}$ condition was designed and validated entirely in the frequency domain.

- **Task 1.** A proportional–derivative attitude law with weak negative drift feedback stabilises the rigid airframe (open-loop pole at +1.84 rad/s) and meets the design targets $|GM| \approx 6$ dB, $|PM| \approx 30^\circ$. The conditional-stability nature of the loop was identified and accounted for in the interpretation of the margins.
- **Task 2.** Introducing the TVC actuator, the transport delay and the lightly damped bending mode destabilises the loop through a +39 dB resonance. A gain-stabilising notch centred on ω_{BM} restores stability while preserving the rigid margins, with negligible change to the nominal gust response. Gain stabilisation was preferred to phase stabilisation because the resonance is too large to be tamed by a phase shift alone.
- **Task 3.** The fixed Task 2 controller is robust to $\pm 30\%$ independent variations of μ_α and μ_c : all four corners remain stable. Aerodynamic uncertainty primarily erodes the rigid-body margin and nominal performance, whereas control-effectiveness uncertainty primarily erodes the bending margin.

The design illustrates the textbook trade-offs of launch-vehicle attitude control [1, 2]: enough gain to stabilise the unstable airframe, but not so much as to excite the structural modes, with the Nichols chart providing a single, unified view of both constraints. A Simulink block-diagram mirror of the closed loop reproduces the script results and is documented in the accompanying `models/SIMULINK_GUIDE.md`.

References

- [1] A. L. Greensite. *Analysis and Design of Space Vehicle Flight Control Systems*. Spartan Books, New York, 1970.
- [2] B. Wie. *Space Vehicle Dynamics and Control*. AIAA Education Series, Reston, VA, 2nd edition, 2008.
- [3] G. F. Franklin, J. D. Powell, and A. Emami-Naeini. *Feedback Control of Dynamic Systems*. Pearson, Upper Saddle River, NJ, 7th edition, 2015.
- [4] A. E. Bryson and Y.-C. Ho. *Applied Optimal Control: Optimization, Estimation and Control*. Taylor & Francis, New York, revised edition, 1975.
- [5] M. Noton. *Modern Control Engineering*. Pergamon Press, Oxford, 1972.